

Trainings ▾

Preparatory Steps

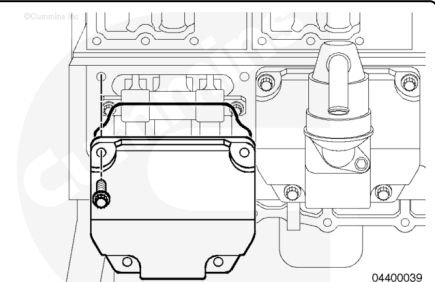
- Remove the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/89/89-003-011.html>) in Section 3.
- Remove the rocker levers. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures/89/89-003-009.html>) in Section 3.
- Remove the push rods. Refer to Procedure 004-014 (</qs3/pubsys2/xml/en/procedures/89/89-004-014.html>) in Section 4.
- Remove the crankcase breather assembly, if necessary. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/89/89-003-001-tr.html>) in Section 3.
- Remove the fuel supply lines, if necessary. Refer to Procedure 006-024 (</qs3/pubsys2/xml/en/procedures/89/89-006-024.html>) in Section 6.
- Remove the fuel drain lines, if necessary. Refer to Procedure 006-013 (</qs3/pubsys2/xml/en/procedures/89/89-006-013.html>) in Section 6.
- Remove the fuel filters, if necessary. Refer to Procedure 006-015 (</qs3/pubsys2/xml/en/procedures/89/89-006-015.html>) in Section 6.
- Remove the alternator, if necessary. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>) in Section 13

Remove

Remove the cam follower cover.

Remove and inspect the gasket.

Discard the gasket if it is damaged.

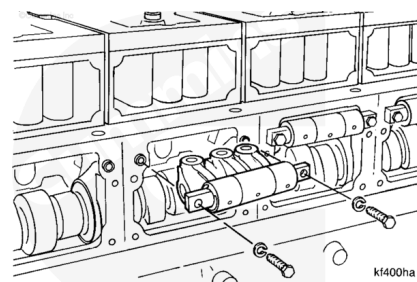


Mark the location of the cam follower assemblies as they are removed, so they can be installed in the original location.

Remove the cam follower assembly capscrews.

Pull the cam follower assembly out until the shaft is off the ring dowels.

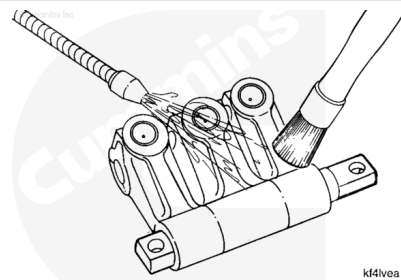
Note : If one of the followers is in the raised position on the cam lobe, it may be necessary to bar the engine over in order to remove the assembly from the block.



Clean and Inspect for Reuse

⚠ WARNING ⚠

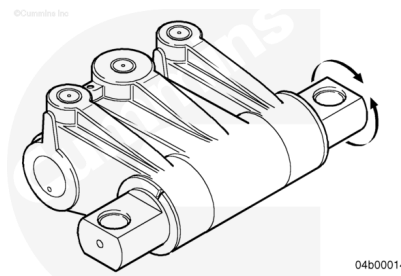
When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use solvent to clean the cam follower assembly.

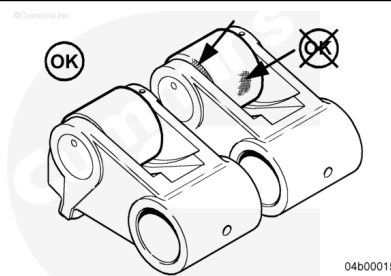
Do **not** touch the machined surfaces of the cam follower roller with bare hands. Use rubber gloves or a similar covering.

The shaft **must** rotate easily. If resistance is apparent, disassemble and check for burrs on the shaft.



Inspect the cam follower lever and roller for damage.

The cam follower lever assembly **must** be replaced if damaged.

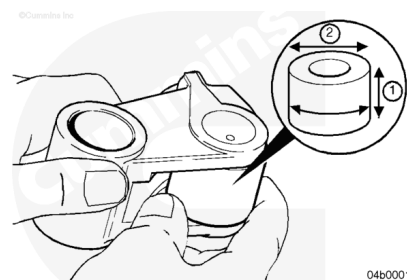


The roller **must** rotate freely.

Measure the side clearance of the roller.

Cam Follower Roller Side Clearance

	mm		in
1	0.075	MIN	0.003
	0.305	MAX	0.012
2	0.082	MIN	0.003
	0.114	MAX	0.005



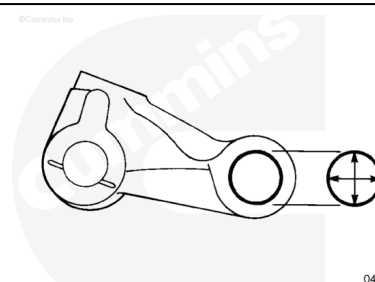
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If the clearances do **not** meet specifications, the cam follower **must** be replaced.

Measure the lever bore side diameter. Replace if worn or damaged.

Lever Bore Inside Diameter

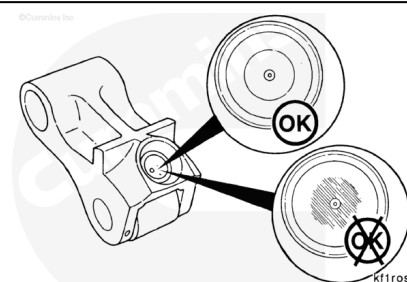
mm		in
25.000	MIN	0.984
25.021	MAX	0.985



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Inspect the push rod socket for an even seating pattern or excessive wear.

When parallel grooves and scratches are observed, or the contact area extends into the oil hole chamfer of the socket, the socket **must** be replaced.

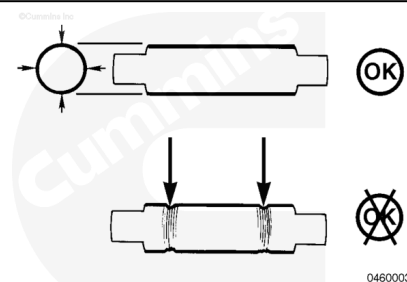


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Check the shaft for damage or roughness. Measure the outside diameter.

Shaft Outside Diameter

mm		in
24.947	MIN	0.982
24.962	MAX	0.983



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Install

Do **not** touch the cleaned machined surfaces of the cam follower roller with bare hands. This can cause rust to form.

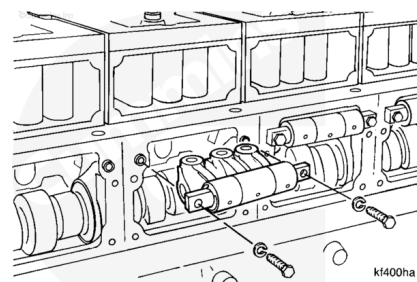
Use clean 15W-40 engine oil to lubricate the camshaft cam follower rollers and cam follower shaft.

Install the cam follower assembly. The shaft **must** fit on ring dowels. The side of the shaft with the ball pressed in **must** face the front of the engine.

Install and tighten the mounting screws.

Torque Value:

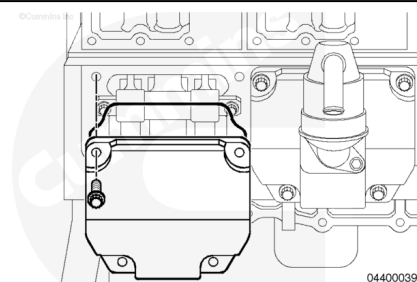
1. 35 n•m [26 ft-lb]
2. 60 n•m [44 ft-lb]
3. 100 n•m [74 ft-lb]



Install the cam follower gasket and cover.

Install the cam follower cover capscrews.

Torque Value: 10 n•m [90 in-lb]



Finishing Steps

- Install the fuel drain lines. Refer to Procedure 006-013 (</qs3/pubsys2/xml/en/procedures/89/89-006-013.html>) in Section 6.
- Install the fuel supply lines. Refer to Procedure 006-024 (</qs3/pubsys2/xml/en/procedures/89/89-006-024.html>) in Section 6.
- Install the crankcase breather assembly. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/89/89-003-001-tr.html>) in Section 3.
- Install the fuel filters. Refer to Procedure 006-015 (</qs3/pubsys2/xml/en/procedures/89/89-006-015.html>) in Section 6.
- Install the alternator. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>) in Section 13.
- Install the pushrods. Refer to Procedure 004-014 (</qs3/pubsys2/xml/en/procedures/89/89-004-014.html>) in Section 4.
- Install the rocker levers. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures/89/89-003-009.html>) in Section 3.
- Adjust the overhead. Refer to Procedure 003-006 (</qs3/pubsys2/xml/en/procedures/89/89-003-006.html>) in Section 3.

- Install the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/89/89-003-011.html>) in Section 3.

Last Modified: 24-Sep-2019
